



in collaboration with



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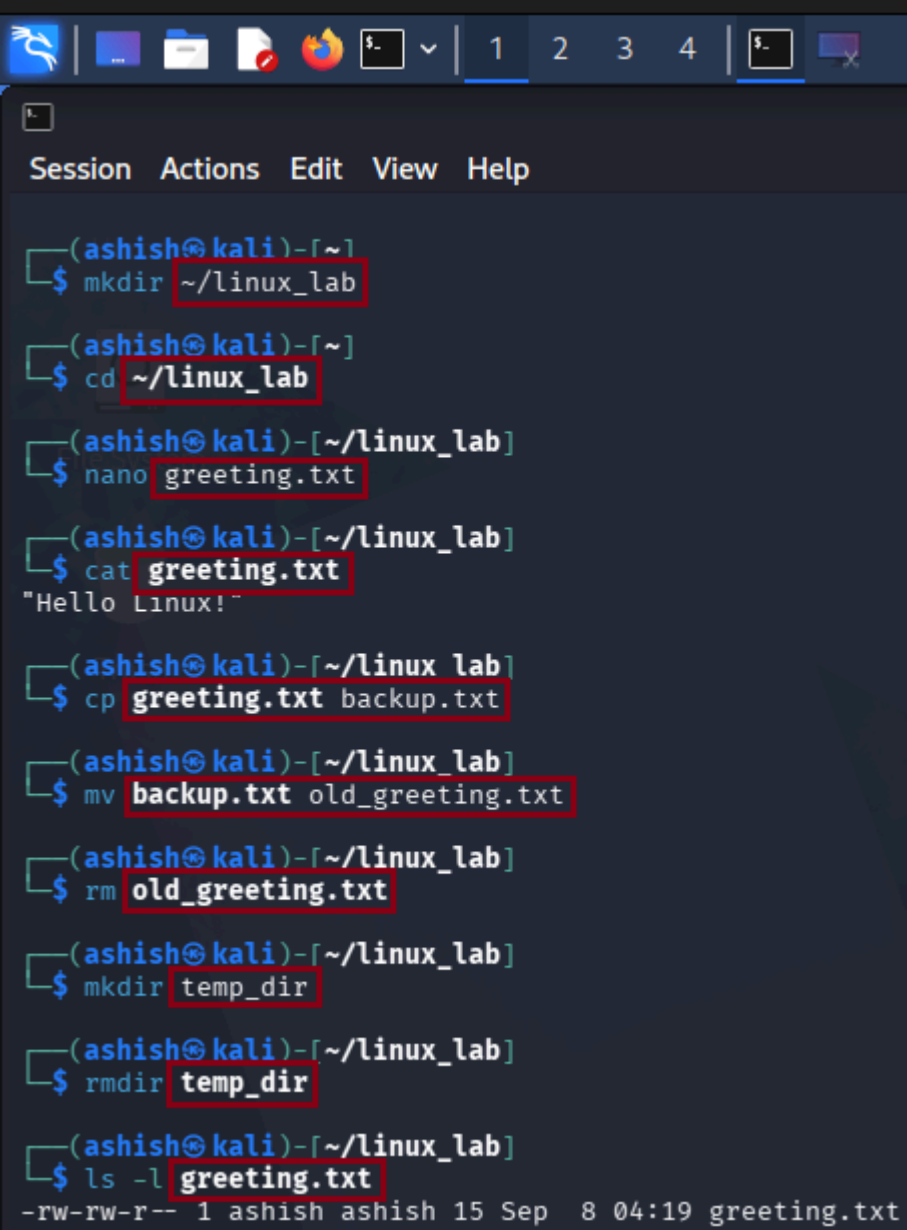
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1.File Operation:

Screen short:

A screenshot of a terminal window with a dark background. The window has a title bar with icons for a terminal, a file manager, a web browser, and a terminal. Below the title bar is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a series of commands and their outputs. The prompt is '(ashish@kali)-[~]'. The commands and outputs are: 1. '\$ mkdir ~/linux_lab' 2. '\$ cd ~/linux_lab' 3. '\$ nano greeting.txt' 4. '\$ cat greeting.txt' followed by the output '"Hello Linux!"' 5. '\$ cp greeting.txt backup.txt' 6. '\$ mv backup.txt old_greeting.txt' 7. '\$ rm old_greeting.txt' 8. '\$ mkdir temp_dir' 9. '\$ rmdir temp_dir' 10. '\$ ls -l greeting.txt' followed by the output '-rw-rw-r-- 1 ashish ashish 15 Sep 8 04:19 greeting.txt'. The file names and paths in the commands are highlighted with red boxes.

```
(ashish@kali)-[~]
$ mkdir ~/linux_lab

(ashish@kali)-[~]
$ cd ~/linux_lab

(ashish@kali)-[~/linux_lab]
$ nano greeting.txt

(ashish@kali)-[~/linux_lab]
$ cat greeting.txt
"Hello Linux!"

(ashish@kali)-[~/linux_lab]
$ cp greeting.txt backup.txt

(ashish@kali)-[~/linux_lab]
$ mv backup.txt old_greeting.txt

(ashish@kali)-[~/linux_lab]
$ rm old_greeting.txt

(ashish@kali)-[~/linux_lab]
$ mkdir temp_dir

(ashish@kali)-[~/linux_lab]
$ rmdir temp_dir

(ashish@kali)-[~/linux_lab]
$ ls -l greeting.txt
-rw-rw-r-- 1 ashish ashish 15 Sep 8 04:19 greeting.txt
```

Fig1: File operation using Linux Command.

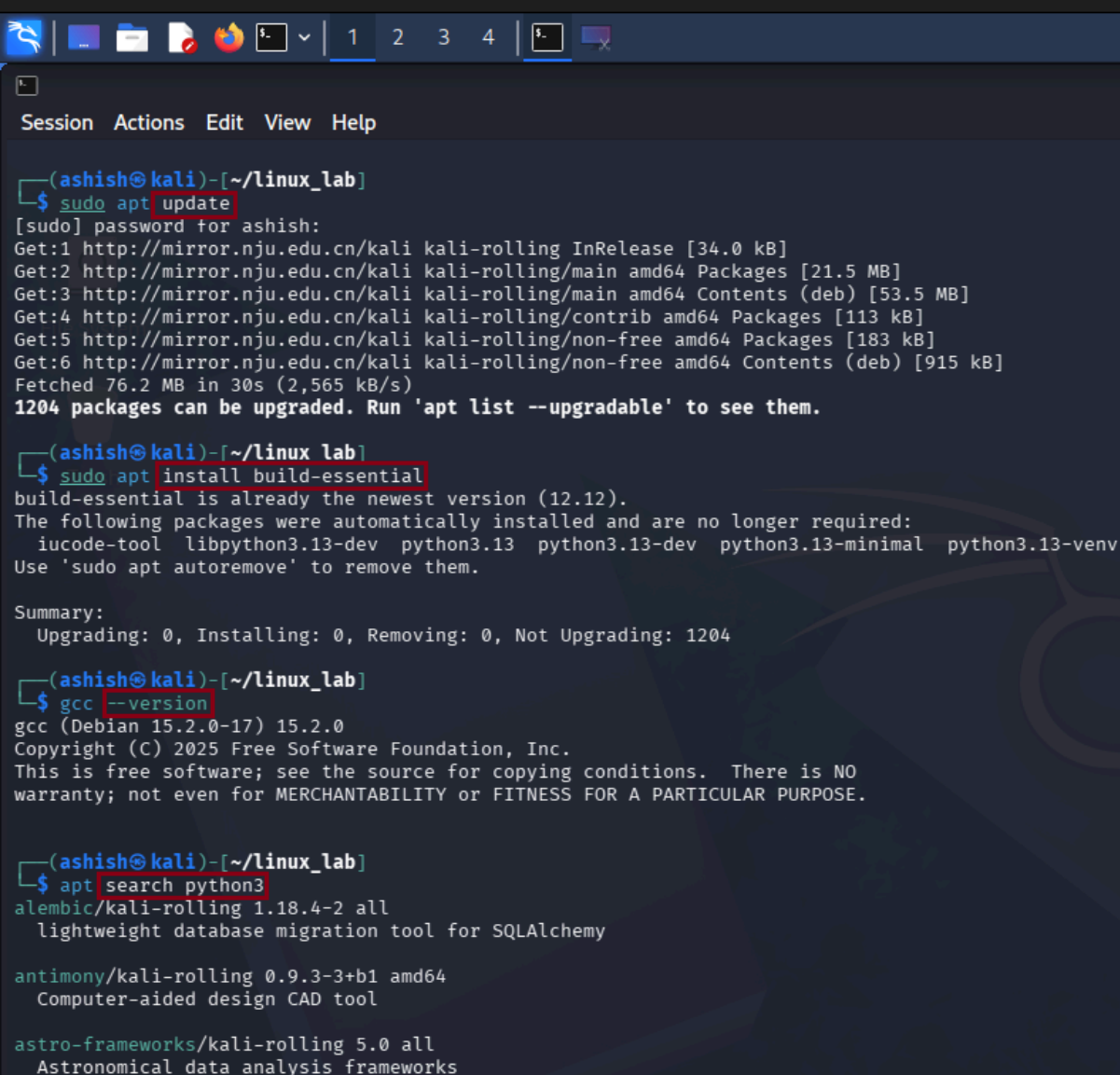
Summary:

The user created a directory called linux_lab, used nano to edit "Hello Linux" into greeting.txt, made a copy of it and then deleted it after renaming, and also created and deleted an empty directory named temp_dir.

They used command ls -l on greeting.txt and found out that the file is still there with permission -rw-rw-r-- and size 15 bytes.

2. APT PACKAGE MANAGEMENT:

Screen short:



```
(ashish@kali)-[~/linux_lab]
$ sudo apt update
[sudo] password for ashish:
Get:1 http://mirror.nju.edu.cn/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirror.nju.edu.cn/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://mirror.nju.edu.cn/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Get:4 http://mirror.nju.edu.cn/kali kali-rolling/contrib amd64 Packages [113 kB]
Get:5 http://mirror.nju.edu.cn/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:6 http://mirror.nju.edu.cn/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.2 MB in 30s (2,565 kB/s)
1204 packages can be upgraded. Run 'apt list --upgradable' to see them.

(ashish@kali)-[~/linux_lab]
$ sudo apt install build-essential
build-essential is already the newest version (12.12).
The following packages were automatically installed and are no longer required:
  iucode-tool libpython3.13-dev python3.13 python3.13-dev python3.13-minimal python3.13-venv
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1204

(ashish@kali)-[~/linux_lab]
$ gcc --version
gcc (Debian 15.2.0-17) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

(ashish@kali)-[~/linux_lab]
$ apt search python3
alembic/kali-rolling 1.18.4-2 all
  lightweight database migration tool for SQLAlchemy

antimony/kali-rolling 0.9.3-3+b1 amd64
  Computer-aided design CAD tool

astro-frameworks/kali-rolling 5.0 all
  Astronomical data analysis frameworks
```

Fig2: APT command.

Summary:

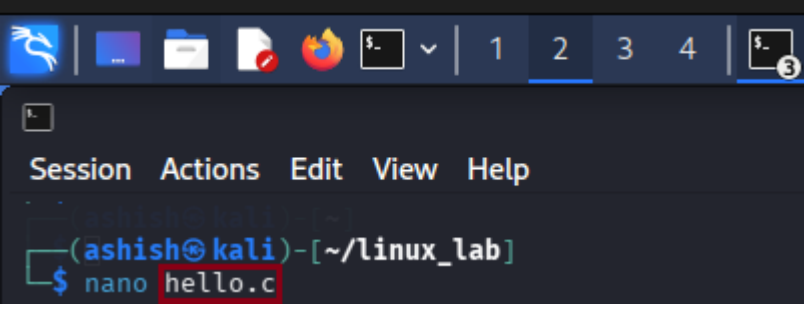
The user executed the command 'sudo apt update' for updating the package list and got 1204 upgradable packages. He installed 'build-essential' that was already there and confirmed GCC version as 15.2.0.

An apt search 'python3' was done that gave a number of astronomy and CAD tools, but was truncated.

3. C COMPILATION:

Screen short:

Step1: Write the code:



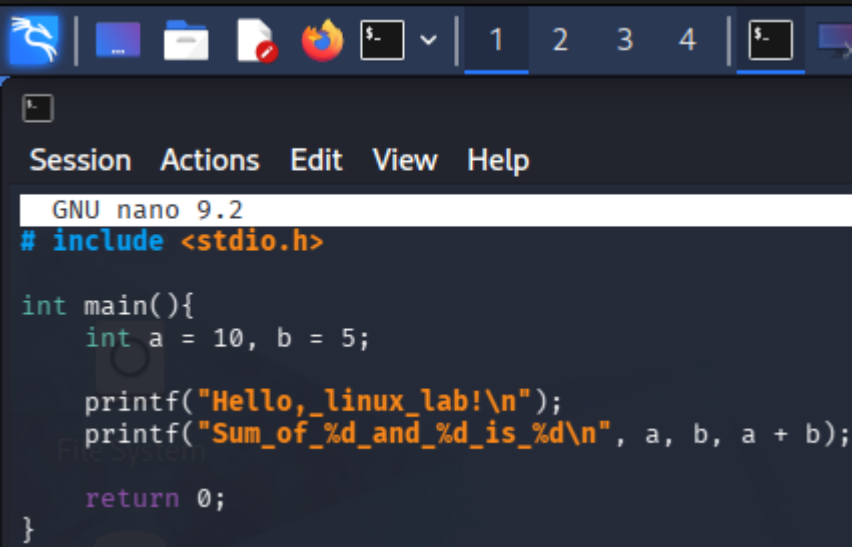
```
(ashish@kali)-[~/linux_lab]
$ nano hello.c
```

Fig3: Create hello.c file.

Summary:

The user launched the nano editor to create or modify a file called hello.c, located in the directory ~/linux_lab, implying that the user is intending to write a program in the C language.

The reason for this assumption comes from the user's prior check of the GCC compiler.



```
GNU nano 9.2
# include <stdio.h>

int main(){
    int a = 10, b = 5;

    printf("Hello, linux_lab!\n");
    printf("Sum_of_%d_and_%d_is_%d\n", a, b, a + b);

    return 0;
}
```

Fig4: C program code.

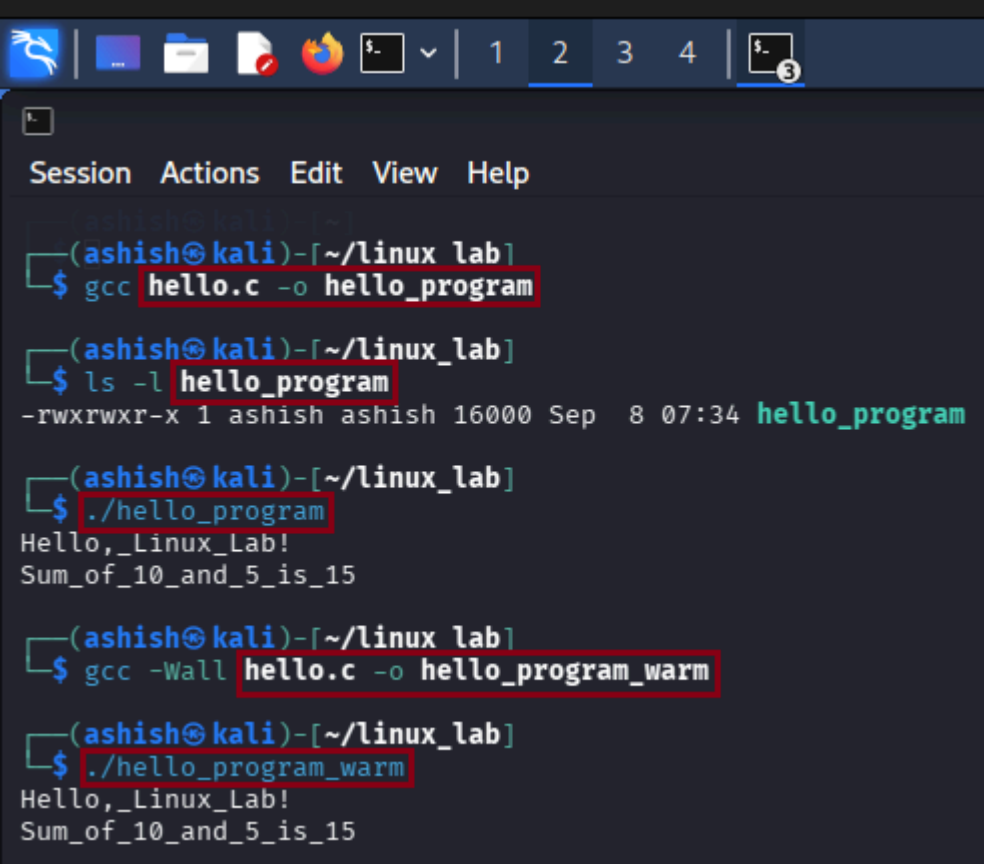
Summary:

The user used nano to edit the hello.c file, and created a simple C program that included the stdio.h library, declared integer a=10 and b=5, printed "Hello, linux_lab!" and the addition of the two numbers using printf().

This is followed by return 0; statement to signify the end of the program which is ready for compilation and execution using GCC.

B. Step 2: Compile and Run:

Screen short:



The screenshot shows a terminal window with a dark background and a menu bar at the top containing 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal prompt is '(ashish@kali)-[~/linux_lab]'. The user enters the command '\$ gcc hello.c -o hello_program', which is highlighted with a red box. The next command is '\$ ls -l hello_program', also highlighted with a red box, showing the output: '-rwxrwxr-x 1 ashish ashish 16000 Sep 8 07:34 hello_program'. The user then runs '\$./hello_program', highlighted with a red box, which outputs 'Hello,_Linux_Lab!' and 'Sum_of_10_and_5_is_15'. The next command is '\$ gcc -Wall hello.c -o hello_program_warm', highlighted with a red box. Finally, the user runs '\$./hello_program_warm', highlighted with a red box, which outputs the same 'Hello,_Linux_Lab!' and 'Sum_of_10_and_5_is_15'.

```
(ashish@kali)-[~/linux_lab]
$ gcc hello.c -o hello_program

(ashish@kali)-[~/linux_lab]
$ ls -l hello_program
-rwxrwxr-x 1 ashish ashish 16000 Sep 8 07:34 hello_program

(ashish@kali)-[~/linux_lab]
$ ./hello_program
Hello,_Linux_Lab!
Sum_of_10_and_5_is_15

(ashish@kali)-[~/linux_lab]
$ gcc -Wall hello.c -o hello_program_warm

(ashish@kali)-[~/linux_lab]
$ ./hello_program_warm
Hello,_Linux_Lab!
Sum_of_10_and_5_is_15
```

Fig5: Compile and run.

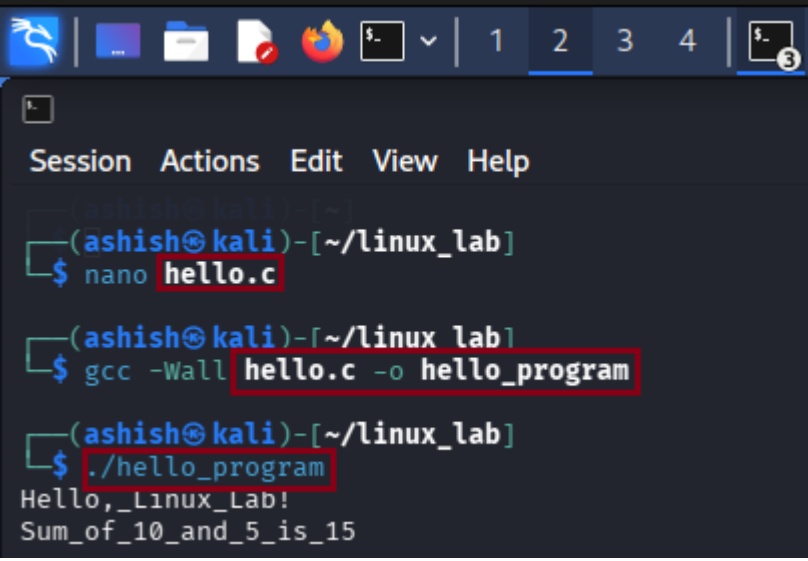
Summary:

The user has compiled hello.c to hello_program through gcc, ran it to print "Hello, Linux_Lab!" and the additional result 15, and then compiled the file using the -Wall option to get the second version called hello_program_warm that prints the same result.

The output of ls -l is showing -rwxrwxr-x for the first file, and there were no problems during the compilations or testing of the files' output.

C. Step 3: Modify and Recompile:

Screen short:



A terminal window with a dark background and a menu bar at the top containing 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a user named 'ashish' on a 'kali' machine in the directory '~/linux_lab'. The user enters three commands, each highlighted with a red box: 'nano hello.c', 'gcc -Wall hello.c -o hello_program', and './hello_program'. The output of the program is displayed below the last command: 'Hello,_Linux_Lab!' and 'Sum_of_10_and_5_is_15'.

```
(ashish@kali)-[~/linux_lab]
$ nano hello.c

(ashish@kali)-[~/linux_lab]
$ gcc -Wall hello.c -o hello_program

(ashish@kali)-[~/linux_lab]
$ ./hello_program
Hello,_Linux_Lab!
Sum_of_10_and_5_is_15
```

Fig6: Modify and recompile.

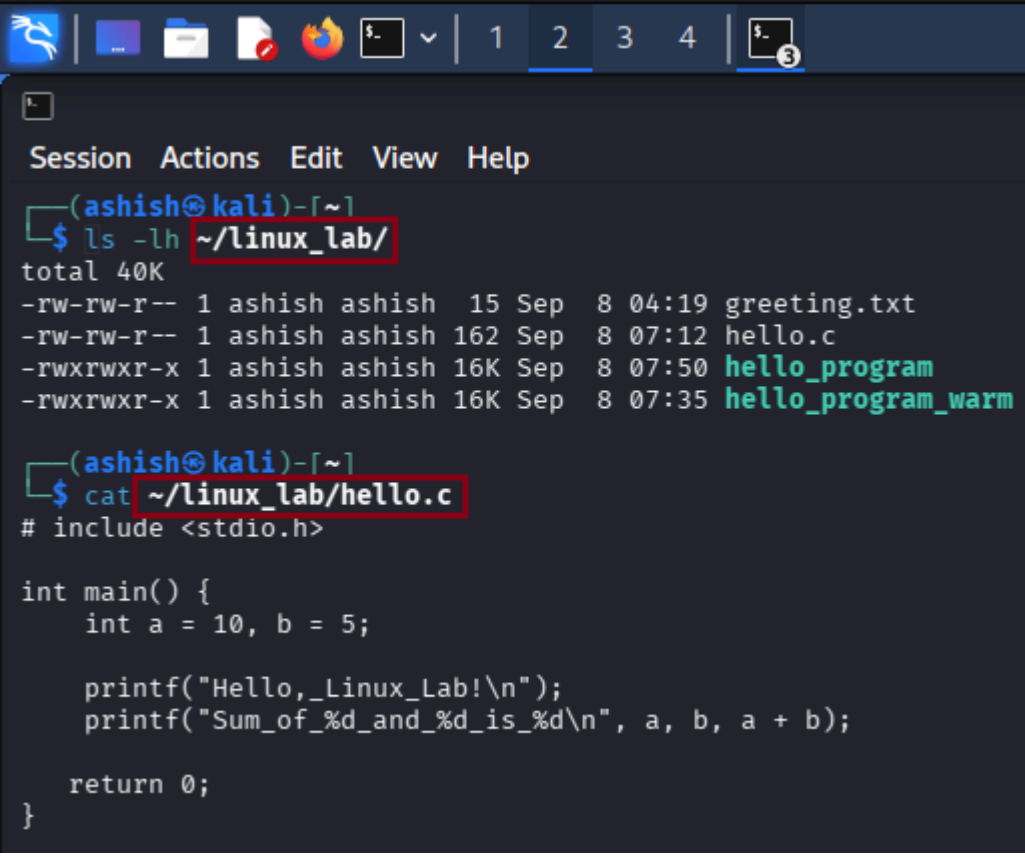
Summary:

The user reopened the file `hello.c` using `nano` (possibly to make changes or review the content), compiled it using `gcc -Wall` into `hello_program` and ran it, outputting "Hello,_Linux_Lab!" and the sum 15.

This is a standard edit-compile-run procedure where the `-Wall` option enables all warnings.

IV. Clean up and Verification:

Screen short:



The screenshot shows a terminal window with a dark background. At the top, there is a window title bar with icons for a terminal, a file manager, a web browser, and a search icon. Below the title bar, there is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal prompt is '(ashish@kali)-[~]'. The first command entered is '\$ ls -lh ~/linux_lab/'. The output shows a list of files: 'greeting.txt', 'hello.c', 'hello_program', and 'hello_program_warm'. The second command entered is '\$ cat ~/linux_lab/hello.c'. The output shows the contents of the file, which includes a C program that prints 'Hello, _Linux_Lab!\n' and calculates the sum of 10 and 5.

```
(ashish@kali)-[~]
$ ls -lh ~/linux_lab/
total 40K
-rw-rw-r-- 1 ashish ashish 15 Sep  8 04:19 greeting.txt
-rw-rw-r-- 1 ashish ashish 162 Sep  8 07:12 hello.c
-rwxrwxr-x 1 ashish ashish 16K Sep  8 07:50 hello_program
-rwxrwxr-x 1 ashish ashish 16K Sep  8 07:35 hello_program_warm

(ashish@kali)-[~]
$ cat ~/linux_lab/hello.c
# include <stdio.h>

int main() {
    int a = 10, b = 5;

    printf("Hello, _Linux_Lab!\n");
    printf("Sum_of_%d_and_%d_is_%d\n", a, b, a + b);

    return 0;
}
```

Fig7: Clean up and verification of code.

Summary:

The user then used the command `ls -lh` to list the files found in `~/linux_lab/` which included `greeting.txt`, `hello.c`, and two executable files, namely `hello_program` and `hello_program_warm`.

The user then used the command `cat` to view the contents of `hello.c` and found that it prints "Hello, Linux Lab!" without the underscores but with the same sum calculation.

Conclusion:

This lab was helpful in developing proficiency in using Linux through tasks such as managing files and directories, updating packages using APT, confirming the GCC compiler, and going through the entire process of C programming including editing, compiling, and execution. The final ls and cat commands showed that the workspace had everything in order, namely, greeting.txt, hello.c, and two binaries.